Récupération, analyse et visualisation de données numériques

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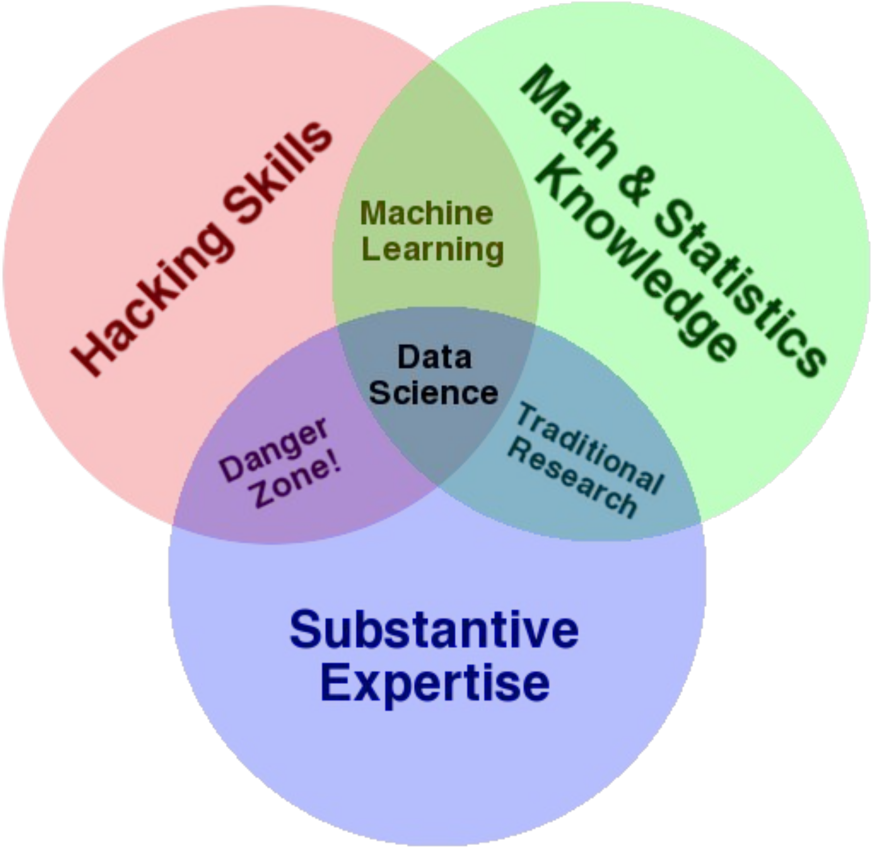

Hacking Skills
Math & Statistics Knowledge
Machine Learning
Data Science
Danger Zone!
Traditional Research
Substantive Expertise

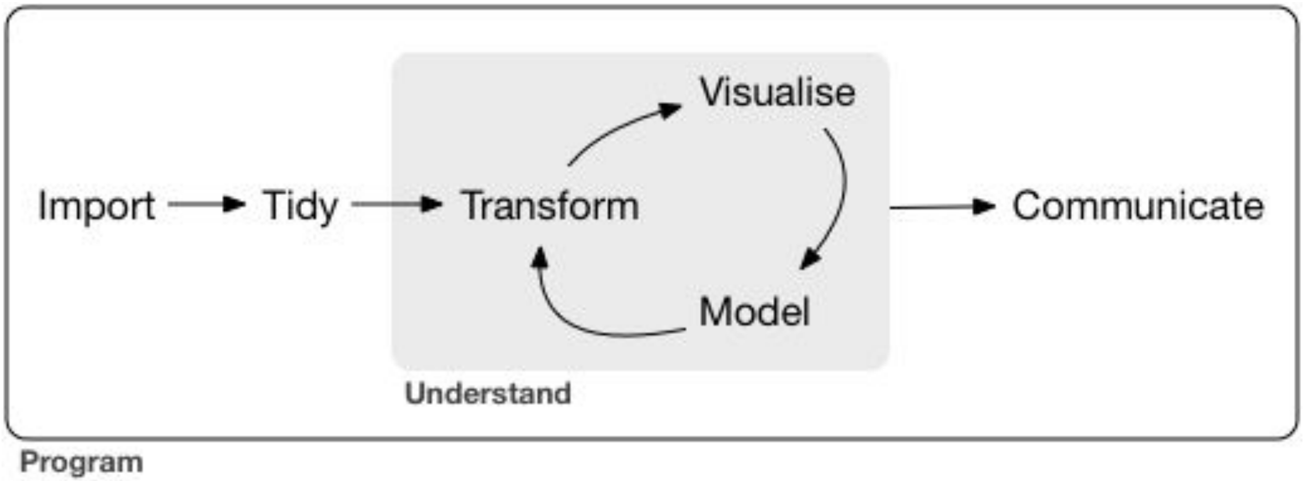
Applied Data Analysis

Applied: nous mettons l'accent sur les techniques et les méthodes dans des scénarios réels, plutôt que sur les mises en œuvre (mais la théorie reste importante !). En d'autres termes, nous nous concentrons sur ce que fait une technique ou une méthode, plutôt que sur la manière précise dont elle fonctionne.

Data: nous utilisons des ensembles de données trop volumineux pour être traités (c'est-à-dire lus) manuellement. Nous considérons également les ensembles de données qui sont trop volumineux pour être consultés manuellement dans leur intégralité sans risque élevé de biais.

Analysis: les données et les outils ne servent pas à grand-chose s'il n'y a pas de motivation, de question ou de besoin d'information, qui devraient, à son tour, aider à créer de nouvelles perspectives ou connaissances.

Comment conceptualiser l'analyse de données ?



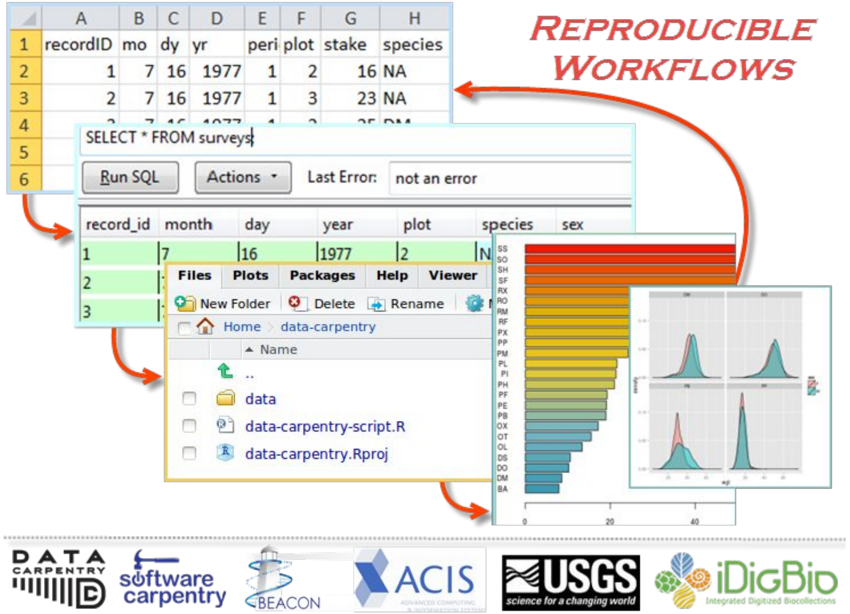
Voir https://r4ds.had.co.nz/introduction.html

Aperçu du cours

Bloc 1	Bloc 2	Bloc 3	Bloc 4	Bloc 5
Introduction et data carpentry	Tidy data	Analyse exploratoire des données	Visualisation des données	Publier données et code, communiquer les résultats

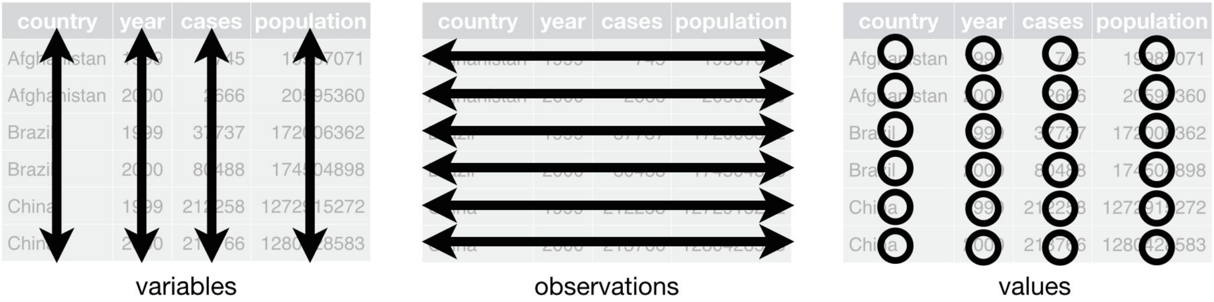
Jeudi 6.3.2025 : journée du CS, GamMAH, 14h-17h

Bloc 1 : Data carpentry



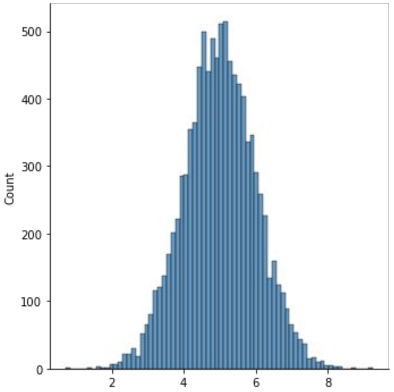
- Maîtriser Pandas
- Web scraping, APIs
- Nettoyage des données
- Importer/exporter les données

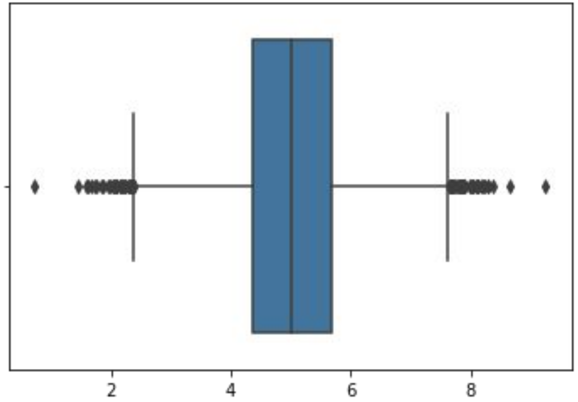
Bloc 2 : Tidy data

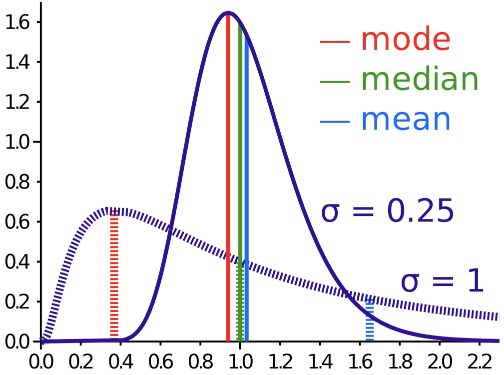


- Comment organiser les données dans une façon « tidy »
- Manipuler des tableaux : les opérations `merge` et `join` (fusionnement)
- Créer des vues sur les données (`group by`)

Bloc 3 : Analyse exploratoire des données

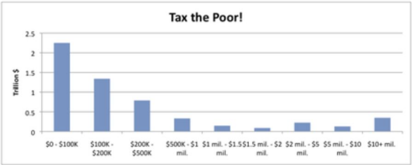


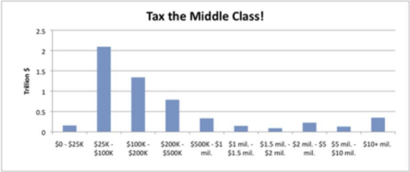


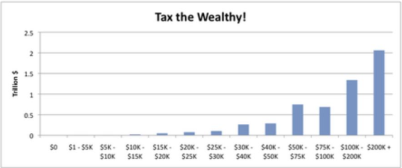


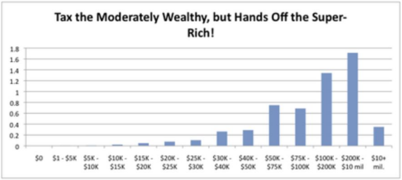
- Distribution des données, valeurs aberrantes (outliers)
- Statistiques descriptives de base (médiane, écart-type)
- Choisir une représentation graphique (plot) en fonction du type des données
- *Plotting* = moyen d’explorer et de mieux connaître un jeu de données
- *Plotting* avec `matplotlib` et `seaborn`

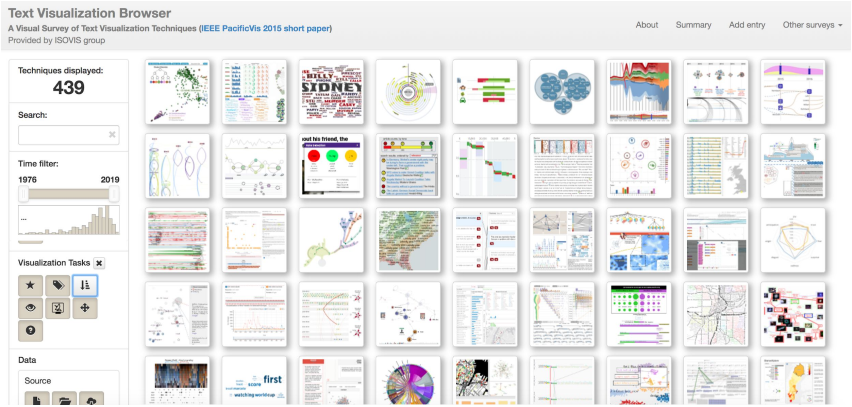
Bloc 4 : Visualisation des données











- Comment mentir avec les données
- Analyse de réseau
- Cartographier les données (geomapping)

Bloc 5 : Publier données et code, communiquer les résultats

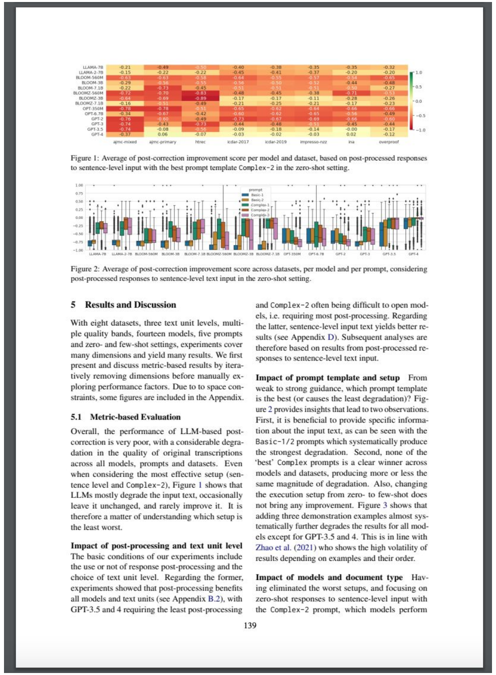


Figure 1: Average of post-correction improvement score per model and dataset, based on post-processed responses to sentence-level input with the best prompt template `Complex-2` in the zero-shot setting.

Figure 2: Average of post-correction improvement score across datasets, per model and per prompt, considering post-processed responses to sentence-level text input in the zero-shot setting.

5 Results and Discussion

With eight datasets, three text unit levels, multiple quality bands, fourteen models, five prompts and zero- and few-shot settings, experiments cover many dimensions and yield many results. We first present and discuss metric-based results by iteratively removing dimensions before manually exploring performance factors. Due to to space constraints, some figures are included in the Appendix.

5.1 Metric-based Evaluation

Overall, the performance of LLM-based post-correction is very poor, with a considerable degradation in the quality of original transcriptions across all models, prompts and datasets. Even when considering the most effective setup (sentence level and `Complex-2`), Figure 1 shows that LLMs mostly degrade the input text, occasionally leave it unchanged, and rarely improve it. It is therefore a matter of understanding which setup is the least worst.

Impact of post-processing and text unit level The basic conditions of our experiments include the use or not of response post-processing and the choice of text unit level. Regarding the former, experiments showed that post-processing benefits all models and text units (see Appendix B.2), with GPT-3.5 and 4 requiring the least post-processing and `Complex-2` often being difficult to open models, i.e. requiring most post-processing. Regarding the latter, sentence-level input text yields better results (see Appendix D). Subsequent analyses are therefore based on results from post-processed responses to sentence-level text input.

Impact of prompt template and setup From weak to strong guidance, which prompt template is the best (or causes the least degradation)? Figure 2 provides insights that lead to two observations. First, it is beneficial to provide specific information about the input text, as can be seen with the `Basic-1/2` prompts which systematically produce the strongest degradation. Second, none of the 'best' `Complex` prompts is a clear winner across models and datasets, producing more or less the same magnitude of degradation. Also, changing the execution setup from zero- to few-shot does not bring any improvement. Figure 3 shows that adding three demonstration examples almost systematically further degrades the results for all models except for GPT-3.5 and 4. This is in line with Zhao et al. (2021) who shows the high volatility of results depending on examples and their order.

Impact of models and document type Having eliminated the worst setups, and focusing on zero-shot responses to sentence-level input with the `Complex-2` prompt, which models perform

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- Bonnes pratiques de codage (modules Python, tests, documentation)
- Où et comment publier un jeu de données
- Comment communiquer les résultats d'une analyse quantitative (bonnes pratiques)

Exemples de jeux de données

- Collection de livres du XIXe siècle de la British LIbrary (métadonnées + texte)
- Contrats d’apprentissage à Venise (XVIe-XVIIe siècle, donnée numérique)
- Corpus français de pièces théâtrales (XVIIe siècle), projet [TextEnt]
- Collection de tweets par Elon Musk (série chronologique)

Évaluation

Projet final :

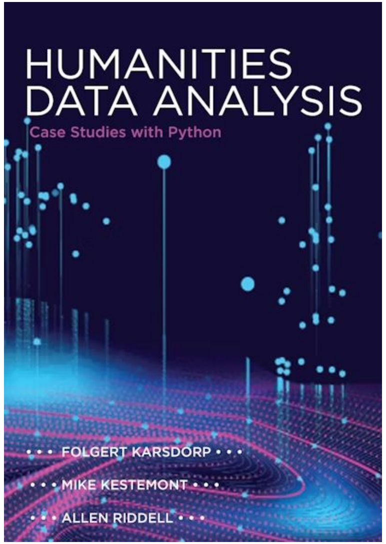
- Groupes de 2-3 personnes max.
- Présentation finale de tous les projets (leçon du 15.5.2025).
- Le projet peut porter sur une ou plusieurs étapes d’une chaîne d’analyse des données (data cleaning, exploratory data analysis, modelling, static/interactive visualizations).
- Jeu de données : a) déjà existant, b) nouveau, c) existant mais jamais publié

Communication :

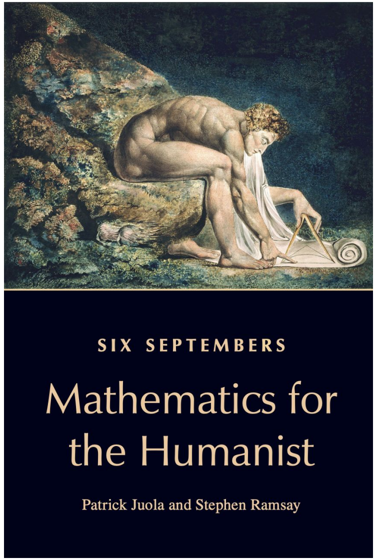
- Canal dédié sur Discord (lien)

Bibliographie

É. Schultz, M. Bussonnier, *Python pour les SHS : Introduction à la programmation pour le traitement de données*, Presses universitaires de Rennes (2021)



F. Karsdorp, M. Kestemont and A. Riddell, *Humanities Data Analysis: Case Studies with Python*,_ Princeton University Press (2021)



P. Juola and S. Ramsay, *Six Septembers: Mathematics for the Humanist* (le chapitre 2 en particulier)

Examples of data analysis applications

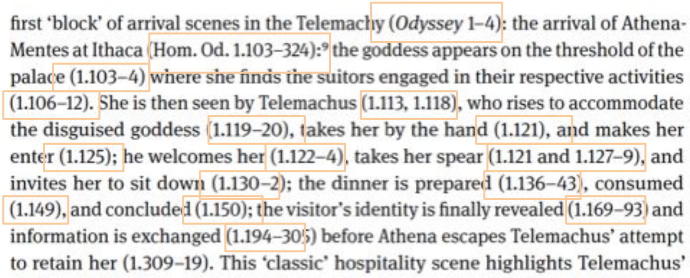
Example of ADA: Mining Classics Citations from JSTOR

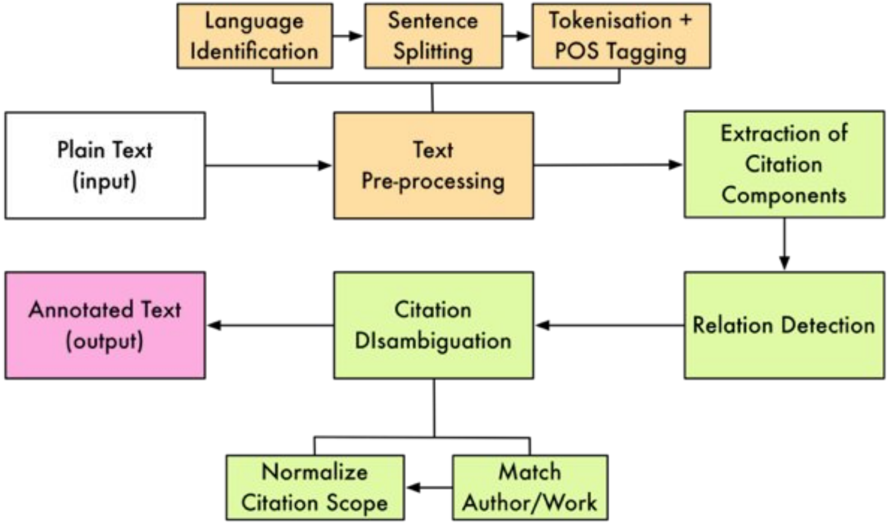
Mining of canonical references from Classics articles in JSTOR, to study scholarly reception of classical texts via citations as a proxy for scholar's attention.

Pre-print: https://doi.org/10.5281/zenodo.3736455

Data processing: mining canonical citations

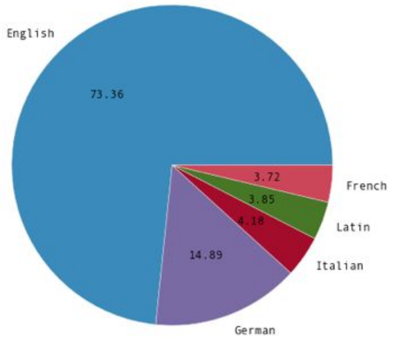
Example of canonical references


first 'block' of arrival scenes in the Telemachy (*Odyssey* 1–4): the arrival of Athena-Mentes at Ithaca (Hom. Od. 1.103–324):[9] the goddess appears on the threshold of the palace (1.103–4) where she finds the suitors engaged in their respective activities (1.106–12). She is then seen by Telemachus (1.113, 1.118), who rises to accommodate the disguised goddess (1.119–20), takes her by the hand (1.121), and makes her enter (1.125); he welcomes her (1.122–4), takes her spear (1.121 and 1.127–9), and invites her to sit down (1.130–2); the dinner is prepared (1.136–43), consumed (1.149), and concluded (1.150); the visitor's identity is finally revealed (1.169–93) and information is exchanged (1.194–305) before Athena escapes Telemachus' attempt to retain her (1.309–19). This 'classic' hospitality scene highlights Telemachus'



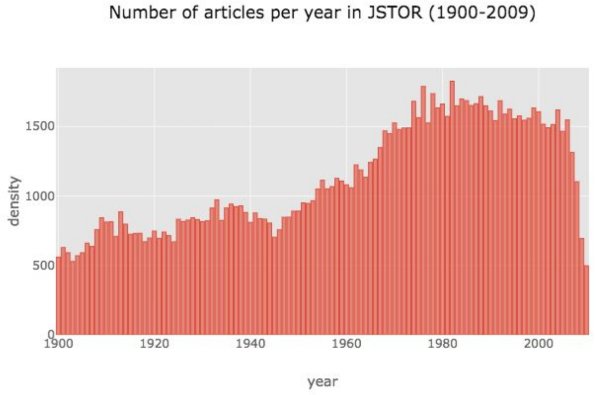
https://citedloci.org/

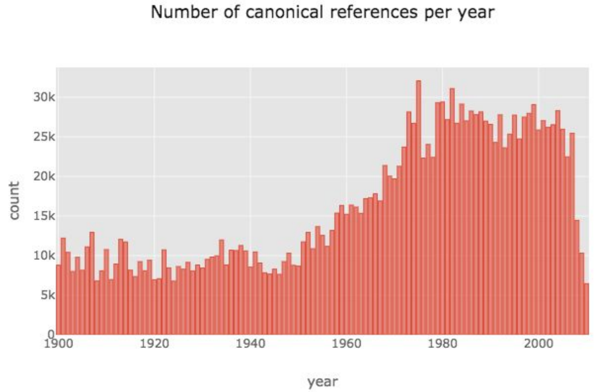
Data overview: Classics articles in JSTOR



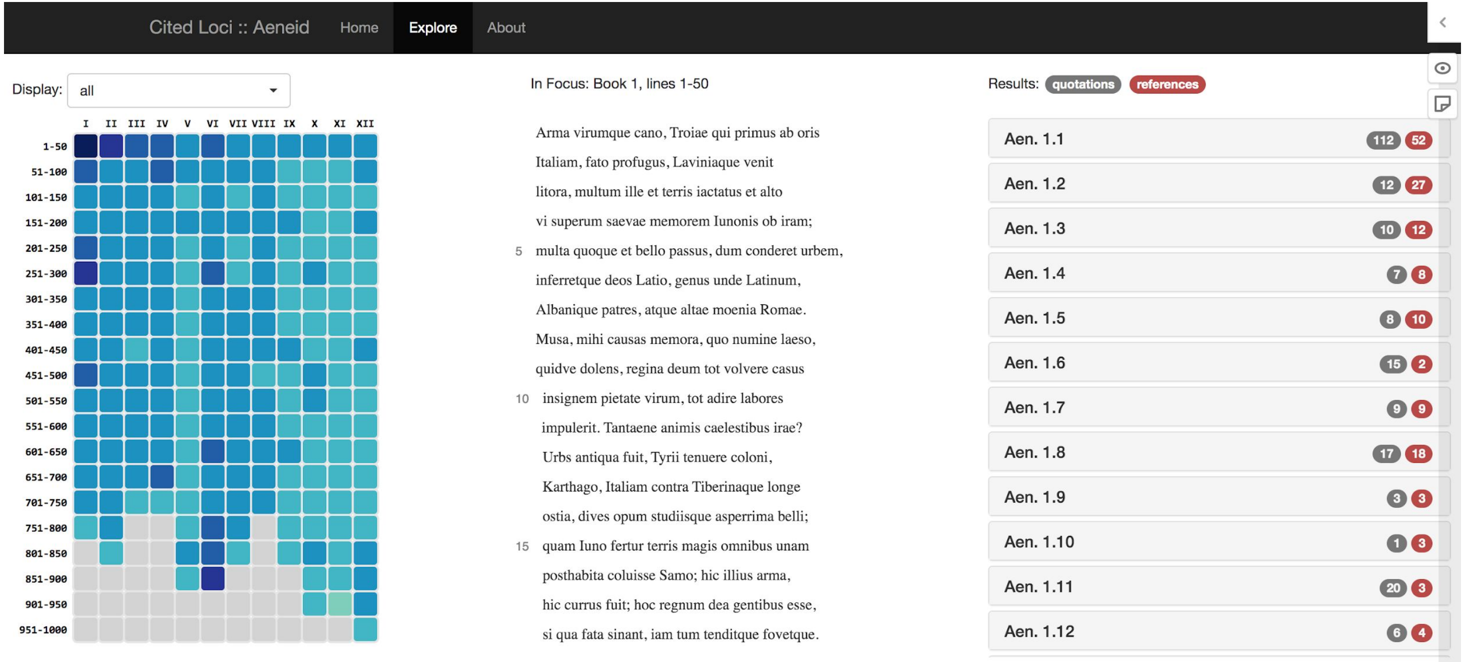
	Number
Total articles	138,821
Successfully processed articles	119,723
Sentences	34,853,399
Tokens	865,075,857
Extracted canonical references	1,649,868
Extracted author mentions	1,448,163
Extracted work mentions	4,665

Table 2: Basic statistics about the JSTOR data.





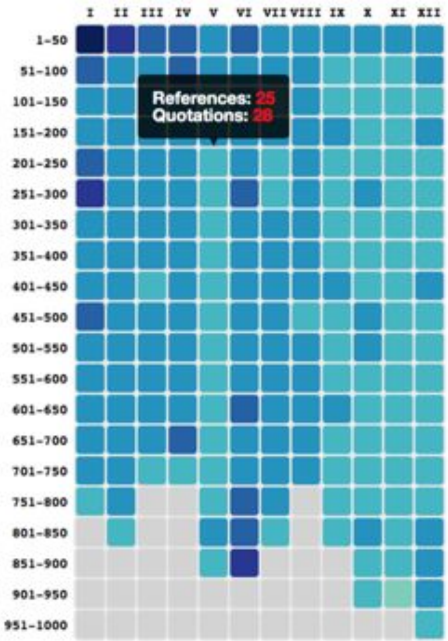
Interactive data visualization: *Aeneid* in JSTOR



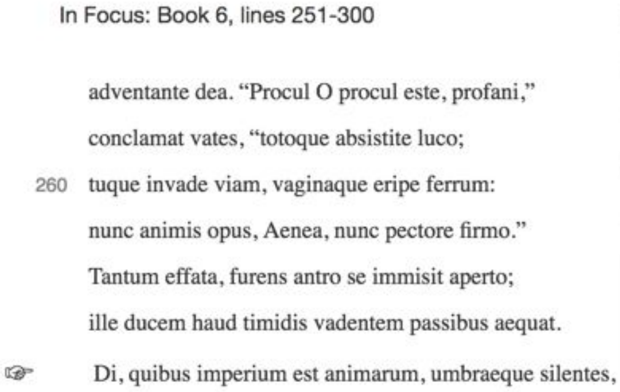
https://mromanello.github.io/Aeneid_in_JSTOR/

http://labs.jstor.org/blog/cited-loci-of-the-aeneid/

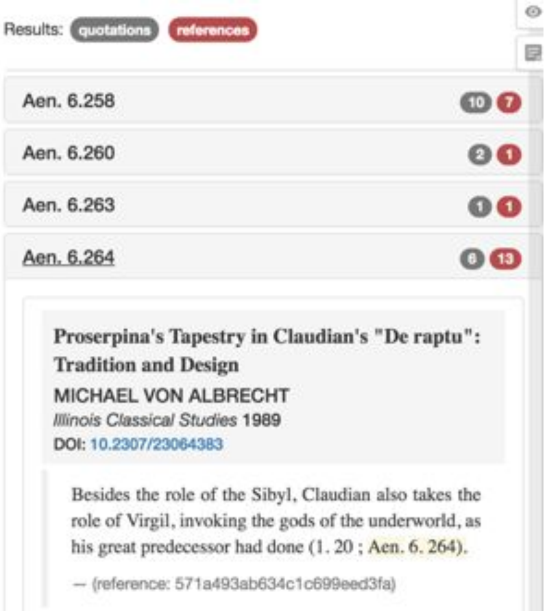
Interactive data visualization: *Aeneid* in JSTOR



Distant reading

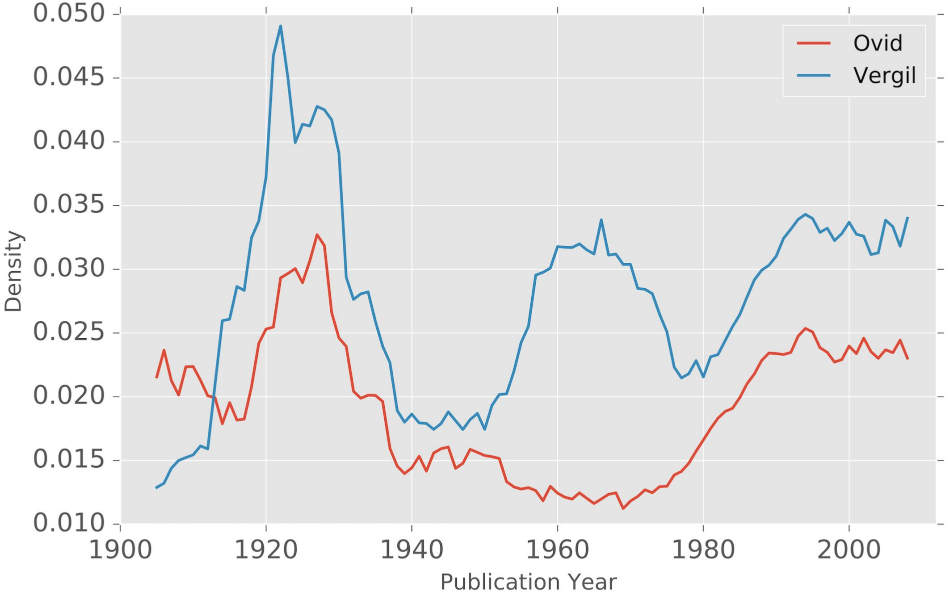


Close reading
(primary literature)



Close reading
(secondary literature)

Longitudinal analysis: Waves of reception in Ovid and Vergil



T. Ziolkowski 2009, *Ovid in the Twentieth Century.*